

Starting at 7:05

Welcome To Advanced Module.

Today's Agenda:-

- ① Prefix Sum Problem
- ② Maximum Subarray Sum.
- ③ Max Absolute Sum.

① Initially all elements of an arr [N] are 0. Then you are given Q queries. Every query contains idx & value. Increment element from i^{th} index to last idx by value. Return final state of the arr.

arr [] : $[0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0]$
 $\quad \quad \quad +4 \ +4 \ +4 \ +4$

Queries

idx value

3 4

1 2

5 -3

5 2

$+2 \ +2 \ +2 \ +2 \ +2 \ +2$

$-3 \ -3$
 $+2 \ +2$

$[0 \ 2 \ 2 \ 6 \ 6 \ 3 \ 3]$
 $\quad \quad \quad 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6$

$\hookrightarrow$ ans.

Brute Force.

① For every query, I am going to iterate from index i to $N-1$.

$\hookrightarrow N$

$$T.C \rightarrow O(N \times A)$$

```

foo(i = 1; i <= A; i++)
{
    idx = a[i]; val = val[i];
    foo(j = idx; j < N; j++)
    {
        arr[j] += val;
    }
}

```

$N \times A$

a_0	a_1	a_2	a_3	a_4	a_5
-------	-------	-------	-------	-------	-------

$prSum[i]$
 a_0 a_0 a_0 a_0 a_0 '

 $+ a_1$ a_1 a_1 a_1 1

 a_2 a_2 a_2

 a_3 a_3

 a_4

$\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \end{matrix}$

 $[0 \quad 2 \quad 0 \quad 4 \quad 0 \quad -3 \quad 0]$

$[0 \quad 2 \quad 2 \quad 6 \quad 6 \quad 3 \quad 3]$

$a[i] \rightarrow idx$

$a[i] \rightarrow \text{value}$

-// take input of all queries.

```
for (i = 0; i < Q; i++)
```

```
{ idx = a[i][0];  
  value = a[i][1];
```

$O(Q)$

```
    arr[idx] += val;
```

```
}
```

// Apply Prefix Sum

```
for (i = 1; i < N; i++)
```

```
{ arr[i] = arr[i] +  
  arr[i-1];
```

```
}
```

```
}
```

```
return arr;
```

T.C $\rightarrow O(N+Q)$

S.C $\rightarrow O(1)$.

H.W.

How my logic changes

if the original array

is non zero.

Variation . . . Google . . .

Now, my query contain
starting index, value AND also an
ending index.

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Queries: $+4 +4 +4 +4 +4 +4$

<u>s.</u>	<u>e</u>	<u>val</u>			
3	6	3			
2	5	4			
5	7	-1			

Final

$$\begin{bmatrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 0 & 0 & 4 & 7 & 7 & 6 & 2 & -1 \end{bmatrix}$$
$$\begin{aligned} a[i][0] &= s; \\ a[i][1] &= e; \\ a[i][2] &= val; \end{aligned}$$
$$\text{for } (\hat{r} = 0; \hat{r} < Q; \hat{r}++)$$

$\{$
 $s = a[i][0];$
 $e = a[i][1];$
 $val = a[i][2];$
 $o(1)$
 $arr[s] += val;$
 $\quad \text{if } (e+1 < N)$
 $\} \quad arr[e+1] -= val;$

// Apply Prefix Sum;

$O(N)$.

T.C $\rightarrow O(Q + N)$.

S.C. $\rightarrow O(1)$.

$arr[8] =$

0	1	2	3	4	5	6	7
0	0	4	3	0	-1	-4	-3

Queries =

0	0	4	7	7	6	2	-17
---	---	---	---	---	---	---	-----

3
6
3

2 5 4
5 7 -1

② Given N array element.
Calculate the max.
Subarray sum.

arr - $\begin{bmatrix} -3 & 2 & 4 & -1 & 3 & -4 & 3 \end{bmatrix}$
= 8 $\rightarrow$ ans.

Idea 1.

Consider all subarrays.
 $\rightarrow O(N^2)$

For every subarray, cal. the
sum & compare it
with the ans. variable

$O(N^2 \times N) \rightarrow O(N^3)$

Idea 2.

Using a carry
forward approach.

ans = $-\infty$ / INT_MIN;

```

for (i = 0; i < N; i++) {
    sum = 0;
    for (j = i; j < N; j++)
        { sum += arr[j];
          ans = max(ans, sum);
        }
    }
return ans;

```

$T.C \rightarrow O(N^2)$
 $S.C \rightarrow O(1)$

Case I :-

All elements in my array are +ve.

3	4	1	6
---	---	---	---

$\hookrightarrow$ the sum of all elements

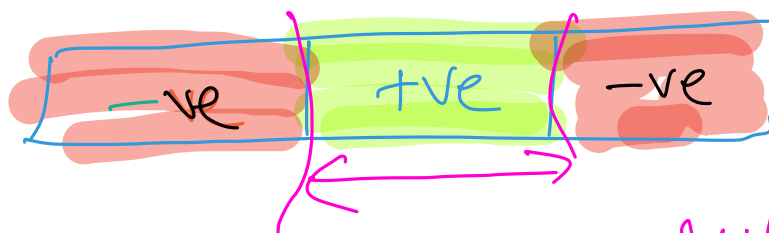
Case 2:-

All element are -ve.

-5 | -3 | -8 | -2

↳ largest element of the array.

Case 3:-



max. subarray sum.



If $sum > 0$; then
carry forward my sum &
update my arr.

arr :

0	1	2	3	4	5	6	7	8	9	10	11
5	6	7	-3	2	-10	-12	8	12	-4	7	-2

70
sum :-

5	11	18	15	17	7	-5 ⁰	8	20	16	23	21
---	----	----	----	----	---	----------------------------	---	----	----	----	----

ans
↓
-∞

5	11	18	-	-	-	-	-	20	-	23	-
---	----	----	---	---	---	---	---	----	---	----	---

Kadane's Algorithm

sum = 0; ans = INT-MIN;

for (i = 0; i < N; i++)

{ sum += arr[i];

ans = max(ans, sum);

if (sum < 0)

sum = 0;

}

// reset

return ans;

}

T.C → O(N)
S.C → O(1)

Q.1

How should I find the indices.



H.W.

Break :- 8:55

Q.1 Max Absolute Difference.

Given an arr. of some integers.
Cal. the max value of $f(i, j)$

$$f(i, j) = |A[i] - A[j]| + |i - j|$$

i, j are indices

⁰
[1 ¹ 3 ² 5]

$|a| = \begin{cases} a; & \text{if } a \geq 0 \\ -a; & \text{if } a < 0 \end{cases}$

i	j	$ A[i] - A[j] + i - j $
0	0	0

$$f(i, i) = f(i, i)$$

$$\text{Ans} = 6.$$

0	1
0	2
1	0
1	1
1	2
2	0
2	1
2	2

$$2 + 1 = 3.$$

$$4 + 2 = 6.$$

$$2 + 1 = 3.$$

$$0$$

$$2 + 1 = 3$$

$$4 + 2 = 6.$$

$$2 + 1 = 3.$$

$$0$$

①

$$f(i, j) = A_i - A_j$$

pick the largest element

a-b.

pick the smallest element

max.

②

$$f(i, j) = (A_i - A_j) + (i - j)$$

$$= (A_i + i) - (A_j + j)$$

Replace each element $\rightarrow$ element + idx.

$$[\overset{0}{8} \quad \overset{1}{2} \quad \overset{2}{7} \quad \overset{3}{4} \quad \overset{4}{-2}]$$

$$= [8 \quad 3 \quad 9 \quad 7 \quad 2]$$

$$= 9 - 2 = 7$$

Maximise the exp.

$$(3) \quad (A[i] - A[j]) + (j - i)$$

$$= (A[i] - i) - (A[j] - j)$$

Replace each element with element - index.

$$= [\overset{0}{8} \quad \overset{1}{2} \quad \overset{2}{7} \quad \overset{3}{4} \quad \overset{4}{-2}]$$

$$= [8 \quad 1 \quad 5 \quad 1 \quad -6]$$

$$= 8 - (-6)$$

$$= 14.$$

Q 2

max.

$$|A[i] - A[j]| + (i - j);$$

$$|a| = \max(a, -a);$$

$$|x - y| = \max(x - y, -(x - y))$$

$$= \max \left[\begin{array}{l} A_i - A_j + i - j; \\ A_j - A_i + i - j; \end{array} \right]$$

$$= \max \left((A_i + i) - (A_j + j); \right. \\ \left. (A_j - j) - (A_i - i); \right.$$

same

$$(A_i - i) - (A_j - j)$$

$$T.C \rightarrow O(\underline{N})..$$

Original Question

maximise $f(i, j)$

$$f(i, j) = f(1, 2);$$

$$= |A[i] - A[j]| \\ + |i - j|$$

$$\begin{array}{cc}
 \underline{A[i]} - A[j] & (i - j) \\
 A[i] - \underline{A[j]} & (j - i)
 \end{array}$$

$= \max$
 Same.

$$\begin{array}{l}
 (A[i] + i) - (A[j] + j) \\
 (A[i] - i) - (A[j] - j) \\
 (A[j] - j) - (A[i] - i) \\
 (A[j] + j) - (A[i] + i)
 \end{array}$$

Code.

① Take max. value of $(A[i] + i) - (A[j] + j)$;

(2)

Take max value of
 $(A[i] - i) - (A[j] - j);$

(3)

Return the greater
of the 2 values.

Code.

// Take max. value of
 $(A_i + i) - (A_j + j).$

max = $-\infty$, min $\rightarrow \infty$

ans = $-\infty$;

for ($i = 0$; $i < N$; $i++$) {

if ($i + arr[i] > \text{max}$)

$O(N)$

$$\{ \max = i + \text{arr}[i]; \}$$

$$\text{if } (i + \text{arr}[i] < \min) \\ \{ \min = i + \text{arr}[i]; \}$$

}

$$\text{ans} = \max(\text{ans}, \max - \min);$$

$$\max = -\infty, \min = \infty$$

$$\text{for } (i = 0; i < N; i++) \{$$

$$\text{if } (\text{arr}[i] - i > \max)$$

$$\{ \max = \text{arr}[i] - i; \}$$

$O(N)$

$$\text{if } (\text{arr}[i] - i < \min) \\ \{ \min = \text{arr}[i] - i; \}$$

}

$$\underline{\max - \min}$$

$ans = \max(ans, \max - \min);$

$\text{return } ans;$

}

$$T.C = \frac{O(N)}{O(1)}$$

Contest is Done.